Ranking Content Refresh Opportunities from Prior Search Signals

FlyRank ML Internship · Refresh / Content Opportunity Scoring · Soyeb Ahmed · 2026-11-01

Abstract

This study asks whether prior search and content signals can rank pages that are more likely to experience a decline in the next 30-day comparison window. It uses the bundled 30,000-row anonymized FlyRank starter dataset and a label based on a greater-than-20% latest-30-day impression decline. A transparent baseline is compared with logistic regression, a decision tree, and a random forest using prior-window and static features under client-holdout validation. The random forest achieved ROC-AUC 0.870 and Precision@50 0.760, versus 0.788 and 0.320 for the baseline. The result is decision support for human review, not a causal estimate or prediction of Google's algorithm.

1. Problem

Prioritize pseudonymized content items for human refresh review when review capacity is limited.

2. Data & leakage control

30,000-row bundled anonymized starter dataset. The model excludes the label itself, latest-window measurements, overlapping 90-day aggregates, current freshness fields, and IDs. It uses prior-30-day signals plus static metadata.

3. Results

Model	ROC-AUC	Avg Precision	Precision@50	Recall	F1
Logistic Regression	0.816	0.695	0.840	0.560	0.616
Decision Tree	0.848	0.698	0.680	0.914	0.749
Random Forest	0.870	0.739	0.760	0.824	0.752
Transparent baseline	0.788	0.598	0.320	—	—

4. Recommendations

The ranked queue supports refresh, expand_and_refresh, review_competitive_position, or monitor actions. The top 50 ranked items had an observed decline rate of 98.0% in this retrospective dataset.

5. Limitations

The run uses the starter slice rather than the gated full warehouse. Client holdout is not a future-time test. Feature importance is associative, not causal. The dataset lacks page text. Nothing here predicts or reverse-engineers Google's ranking algorithm.

6. Reproducibility

Run: pip install -r requirements.txt; python work/scripts/build_capstone.py. Seed: 42. Metrics: work/outputs/capstone_metrics.json. Notebook: work/notebooks/capstone.ipynb.

7. Acknowledgments & data credit

Built on the FlyRank ML Internship dataset. https://flyrank.ai